ModelIt!

Teacher's Guide

Can Renewable Energy Replace Fossil Fuels?

Modeling Energy Systems, Grid Infrastructure, and the Clean Energy Transition

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-2

Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate competing energy transition pathways using model evidence about cost, reliability, carbon reduction, and storage requirements to construct arguments for optimal energy system designs.
Disciplinary Core Idea	**ESS3.A Natural Resources** — All forms of energy production have associated costs, risks, and benefits. The transition from fossil fuels to renewable energy requires evaluating these trade-offs across multiple dimensions including environmental impact, reliability, cost, and equity.
Crosscutting Concept	**Energy and Matter** — Students track how energy flows through the electrical grid system — from generation through transmission, storage, and consumption — and analyze how the constraints on energy conversion and storage determine the feasibility of renewable energy transitions.

CAST Testing Alignment

MULTIPLE CHOICE:

A grid operator increases solar generation capacity from 20% to 50% of total capacity without adding energy storage. What is the most likely immediate outcome? A) Carbon emissions drop by 30% B) Electricity costs decrease C) Grid reliability decreases during evening hours when solar drops to zero and storage cannot fill the gap D) Fossil fuel plants shut down automatically

CONSTRUCTED RESPONSE:

Using your energy system model, explain why achieving the last 20% of a 100% renewable grid (going from 80% to 100%) is significantly more difficult and expensive than the first 20% (going from 0% to 20%). Reference at least three components including storage, grid reliability, and capacity factor in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The global energy system is at an inflection point. Renewable energy — particularly solar and wind — has become the cheapest source of new electricity generation in most of the world, with solar costs falling 90% since 2010 and wind costs falling 70%. Yet fossil fuels still provide approximately 80% of global primary energy. The gap between economic opportunity and actual deployment reveals that the energy transition is not primarily a technology or cost problem — it is a systems integration, infrastructure, and political economy challenge.

The Renewable Cost Revolution

The cost decline of renewable energy has been one of the most dramatic technological transitions in history. Solar photovoltaic costs have fallen from $76 per watt in 1977 to less than $0.20 per watt today — a 99.7% decline. Unsubsidized solar and wind are now cheaper than new coal or gas power plants in regions representing 90% of global electricity generation. Battery storage costs have fallen 97% since 1991 and continue declining at 15-20% per year. These cost curves show no signs of flattening, suggesting that the economic case for renewable energy will only strengthen. However, the cost of generating a kilowatt-hour is different from the cost of delivering reliable power 24/7/365 — the system costs of integration, storage, and transmission add significantly to the generation-only price comparison.

The Intermittency Challenge

Solar panels generate electricity only during daylight hours and at reduced capacity under clouds. Wind turbines generate only when wind speeds fall within their operating range (typically 3-25 meters per second). This intermittency creates a fundamental mismatch between when renewable energy is produced and when consumers need it. In California, the famous duck curve shows excess solar generation during midday that must be curtailed (wasted) because storage is insufficient, followed by a steep evening ramp when solar disappears and demand peaks. Managing this variability requires a portfolio approach: geographic diversity (wind blows somewhere even when calm locally), temporal diversity (solar and wind complement each other), energy storage (batteries for hours, pumped hydro for days), demand flexibility (shifting industrial loads to match generation), and grid interconnection (importing power from regions with surplus generation).

Grid Infrastructure: The Hidden Bottleneck

The electrical grid is the largest machine ever built — a complex system of generation, transmission, distribution, and control that must balance supply and demand every second of every day. The existing grid was designed for centralized fossil fuel plants, not distributed renewable generation. Transitioning to high-renewable operation requires massive infrastructure upgrades: new high-voltage transmission lines to connect remote solar and wind resources to demand centers (the best solar is in deserts, the best wind is offshore or in rural areas), smart grid technology to manage variable generation, bidirectional power flow to accommodate rooftop solar feeding back into the grid, and new frequency regulation techniques to replace the physical inertia lost when spinning fossil fuel generators are retired. These grid upgrades represent a multi-trillion-dollar investment that is technically feasible but requires sustained political will and capital allocation.

Energy Justice and Equitable Transition

The clean energy transition must address equity or it will fail. Low-income communities and communities of color currently bear disproportionate health burdens from fossil fuel pollution — coal plants, refineries, and gas infrastructure are concentrated in disadvantaged neighborhoods. The transition must ensure that these communities benefit first from clean energy, not last. Energy affordability is also critical: if the transition raises electricity costs, low-income households who spend a larger percentage of income on energy will be harmed. A just transition also requires supporting workers and communities dependent on fossil fuel industries through retraining, economic diversification, and direct investment. History shows that ignoring equity in major economic transitions creates political backlash that can derail progress for everyone.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Total Energy Demand, Renewable Generation Capacity, Fossil Fuel Generation, Energy Storage Capacity, Grid Reliability Index, Carbon Emission Rate, and Energy Cost Per Kilowatt-Hour as key components of the energy system.

E **Establish Relationships**

Students determine that renewable capacity alone is insufficient without storage, that intermittency creates a reliability ceiling that storage and grid modernization must address, and that the integrated approach outperforms any single-component strategy.

V **Visualize & Model**

Students build a computational model of a regional energy grid showing how generation, storage, and demand interact to determine reliability and carbon emissions under different renewable penetration levels.

E **Evaluate Outcomes**

Students run rapid deployment, storage breakthrough, and integrated transition scenarios to test predictions about the technical requirements for high-renewable grid operation.

R **Revise & Extend**

Students add nuclear baseload, EV grid integration, or hydrogen storage to explore how additional technologies affect the feasibility and cost of 100% clean energy systems.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Renewable energy is now the cheapest electricity in human history. So why does fossil fuel still power 80% of the world? Today we model the system to find out."*

Watch For

Show cover image. Quick poll: How quickly do you think the world could switch to 100% renewable energy? 5 years? 20 years? Never? Take predictions and reasoning.

Slide 2: Learning Objectives

3 min

Say: *"We\'re not just learning about solar panels today — we\'re modeling the entire energy system to understand why the cheapest technology doesn\'t always win, and what it takes to make it work."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: intermittency — the sun goes down, the wind stops. How do you keep the lights on? This is the central challenge.

Slide 3: Big Question

3 min

Say: *"If renewables are cheaper, why aren\'t they everywhere? The answer is more interesting than you think — it\'s not about the energy, it\'s about the system."*

Watch For

Quick-write: Students write their hypothesis about what barriers prevent faster renewable adoption. Save for comparison after modeling.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll build a model of a regional energy grid. This model will reveal why adding solar panels isn\'t enough — and what else the system needs to work reliably."*

Watch For

Preview LEVER steps. Emphasize: the model will challenge the idea that the energy transition is simple. It\'s an engineering systems challenge of enormous complexity.

Slide 5: Activity 1: Components

8 min

Say: *"Which are external inputs to the energy system, and which are internal system responses? Think about what society demands from the grid versus what the grid produces internally."*

Watch For

Guide sorting: External = Total Energy Demand (consumer behavior), Renewable Generation Capacity (installed infrastructure). Internal = the five system response variables. Discuss why Fossil Fuel Generation is internal — it responds to fill gaps when renewables fall short.

Slide 6: Activity 2: Connections

8 min

Say: *"When Renewable Generation is high during the day but demand peaks in the evening, what happens to Grid Reliability without storage? Trace the pathway from generation to reliability to the role fossil fuels still play."*

Watch For

Students draw arrows. Key discovery: the central role of Energy Storage as the bridge between intermittent generation and constant demand. Without storage, renewable penetration hits a ceiling. Ask: What is that ceiling and why?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios. Pay particular attention to what happens to Grid Reliability in the renewables-without-storage scenario. Then see how the integrated approach changes everything."*

Watch For

Students run scenarios and record observations. Key discussion: Why does adding more solar panels actually decrease grid reliability past a certain point without storage? How does the integrated approach solve this?

Slide 8: Discoveries

5 min

Say: *"The cheapest electron is renewable. But the cheapest reliable grid requires the whole system working together — generation, storage, transmission, and flexibility. Let\'s discuss what this means for energy policy."*

Watch For

Class discussion. Key question: If the technology exists and the economics work, what is actually preventing the transition? Guide toward political economy, infrastructure investment, and fossil fuel industry resistance.

Slide 9: STEM Challenge

12 min

Say: *"Your state has committed to 100% clean electricity by 2045. Design the system — not just the generation, but the storage, grid upgrades, and demand flexibility that make it reliable. Use your model data."*

Watch For

Groups design complete energy system plans with generation mix, storage capacity, grid infrastructure, and equity provisions. Present designs with evidence from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Total Energy Demand, Renewable Generation Capacity
- **INTERNAL:** Fossil Fuel Generation, Energy Storage Capacity, Grid Reliability Index, Carbon Emission Rate, Energy Cost Per Kilowatt-Hour

Relationships

- **Renewable Generation Capacity to Carbon Emission Rate:** NEGATIVE (-) — More renewable generation displaces fossil fuel generation, directly reducing carbon emissions. Each kilowatt-hour of solar or wind that replaces coal or gas eliminates the associated CO2 emissions.
- **Renewable Generation Capacity to Grid Reliability Index:** POSITIVE (+) then NEGATIVE (-) — Initially, adding renewables improves reliability by diversifying the generation mix. Beyond approximately 40% penetration, intermittency begins degrading reliability unless storage and grid flexibility increase proportionally.
- **Energy Storage Capacity to Grid Reliability Index:** POSITIVE (+) — Storage bridges the gap between intermittent renewable generation and constant demand, absorbing excess generation during peak production and dispatching it during low generation, directly improving reliability.
- **Fossil Fuel Generation to Carbon Emission Rate:** POSITIVE (+) — Fossil fuel combustion is the primary source of energy-sector carbon emissions. Each megawatt-hour of coal generation produces approximately 1 ton of CO2; natural gas produces approximately 0.5 tons.
- **Grid Reliability Index to Energy Cost Per Kilowatt-Hour:** NEGATIVE (-) — Maintaining high grid reliability during the transition requires investment in storage, grid infrastructure, and backup capacity that increases system costs. The most reliable configurations cost more than least-cost generation-only calculations suggest.

Key Discoveries

- Adding renewable generation capacity without proportional storage creates grid instability — the model shows that above approximately 40% renewable penetration, intermittency causes frequency instability and curtailment (wasting excess generation) unless storage and demand flexibility increase
- Energy storage is the keystone technology of the renewable transition — the model shows that tripling storage capacity has a larger positive impact on Grid Reliability and Carbon Emission Rate than doubling renewable generation alone
- The cheapest kilowatt-hour is renewable, but the cheapest reliable grid requires the integration of generation, storage, transmission, and demand flexibility — no single component can achieve the transition alone
- The integrated transition scenario achieves 80% renewable operation with maintained grid reliability and 70% carbon reduction — demonstrating that the technical barriers to clean energy are solvable but require coordinated investment in all system components simultaneously

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Renewable energy is unreliable because the sun doesn\'t always shine and the wind doesn\'t always blow"	While individual solar panels and wind turbines are intermittent, a well-designed renewable energy system achieves high reliability through diversification, storage, and grid management. Geographic diversity means wind is almost always blowing somewhere across a large region. Solar and wind are complementary (solar peaks midday, wind often peaks at night). Energy storage bridges generation gaps. Modern weather forecasting allows grid operators to predict renewable output days in advance. Grids in regions like Denmark (80%+ wind) and South Australia (70%+ renewable) demonstrate that high-renewable operation is compatible with reliability.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We need fossil fuels for baseload power that renewables cannot provide"	The concept of baseload power — large plants running constantly — is an artifact of the old grid design, not a physical necessity. Modern grids can provide constant, reliable power from a combination of variable renewables, storage, demand flexibility, and a small amount of dispatchable clean generation. The concept of baseload is being replaced by flexibility — the ability to match generation to demand in real time using diverse sources and storage. The model shows that storage and grid management can replace the role fossil fuel baseload plants currently play.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The energy transition will make electricity unaffordable"	Renewable generation is already the cheapest electricity source in most of the world. While the transition requires significant investment in storage and grid infrastructure, the long-term cost trajectory favors renewables because they have zero fuel costs — once built, the marginal cost of solar and wind electricity is essentially free. Fossil fuel costs are volatile and include massive hidden costs: air pollution causes $2.8 trillion in health damage annually, and climate change costs are projected at $38 trillion per year by 2050. When these externalities are included, the renewable transition saves money overall. The key is ensuring that transition costs are shared equitably rather than falling on those least able to pay.